

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: MLA0301

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

***CO2:** Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Dr G.A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A
Application - Based Questions

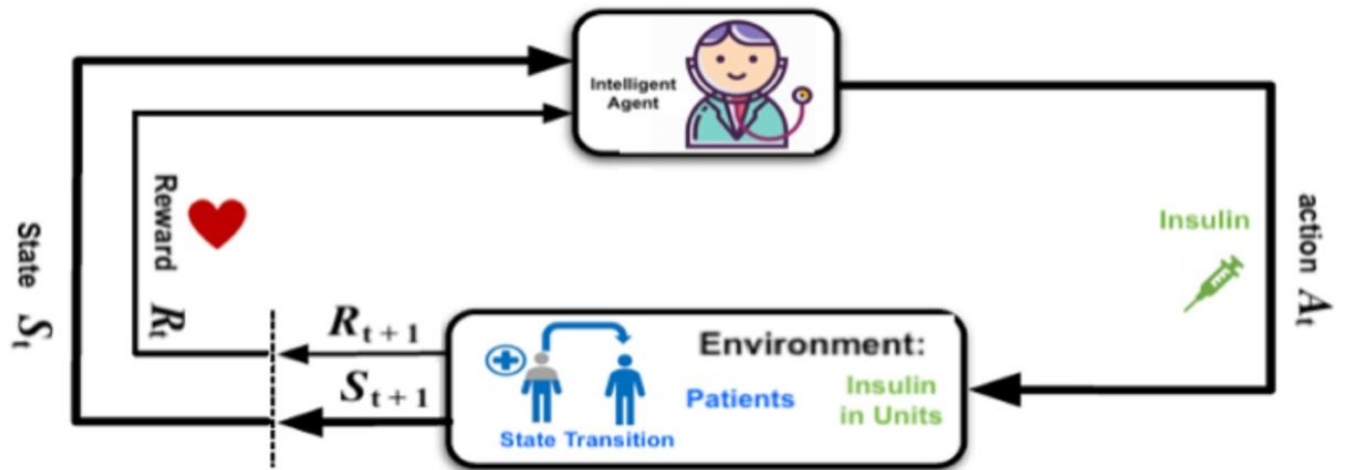
Duration: 2Hrs

1.A healthcare monitoring system uses Reinforcement Learning to adjust medication dosages. Apply RL concepts to define states, actions, and rewards, and explain how learning improves outcomes.

Healthcare Monitoring System using Reinforcement Learning

1. Objective

The objective is to **select an appropriate medication dosage** based on the patient's current health condition while minimizing side effects and improving treatment outcomes.



2. State

The **state** represents the patient's current condition.

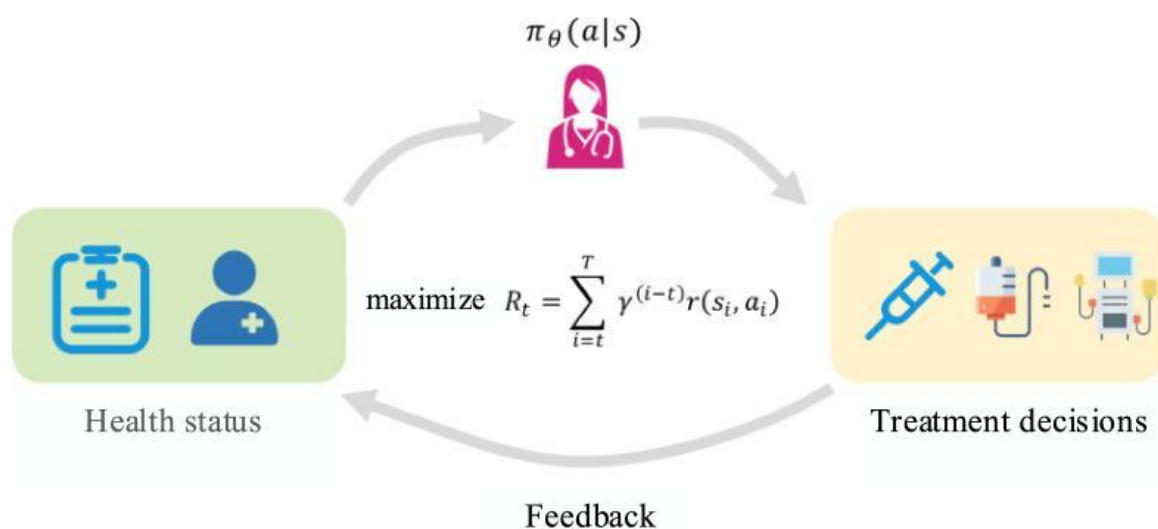
Examples:

- Blood pressure
- Heart rate
- Blood glucose level
- Patient symptoms
- Previous medication dosage
- Side-effect information

3. Actions

The RL agent can select different medication decisions:

- Increase dosage
- Decrease dosage
- Maintain current dosage
- Change medication
- Delay medication



4. Reward

The reward encourages safe and effective treatment.

Positive rewards:

- Patient's health improves
- Target health range is maintained
- Symptoms decrease

Negative rewards:

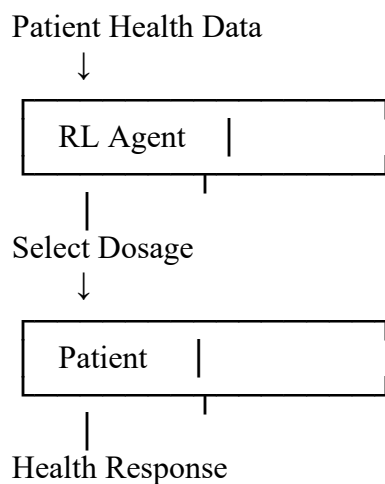
- Severe side effects
- Health condition worsens
- Dosage becomes unsafe

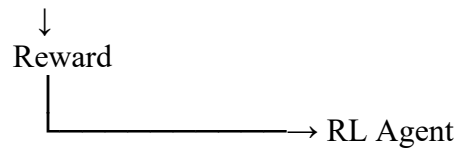


5. How RL improves outcomes

The agent observes the patient's state, chooses a dosage, receives feedback from the resulting health condition, and updates its policy. Over repeated interactions, it learns which dosage decisions provide better outcomes.

Simple Diagram





Conclusion: RL can learn personalized treatment strategies by continuously using patient responses as feedback. In real healthcare, such systems would require strict clinical oversight and safety constraints.

2.Design a model drone delivery system uses Reinforcement Learning for navigation. Recall how exploration and exploitation strategies influence route optimization.

Drone Delivery System using Reinforcement Learning

1. Objective

The objective is to find a **safe and efficient route** for the drone to deliver packages while minimizing travel time, energy consumption, and obstacles.



2. State

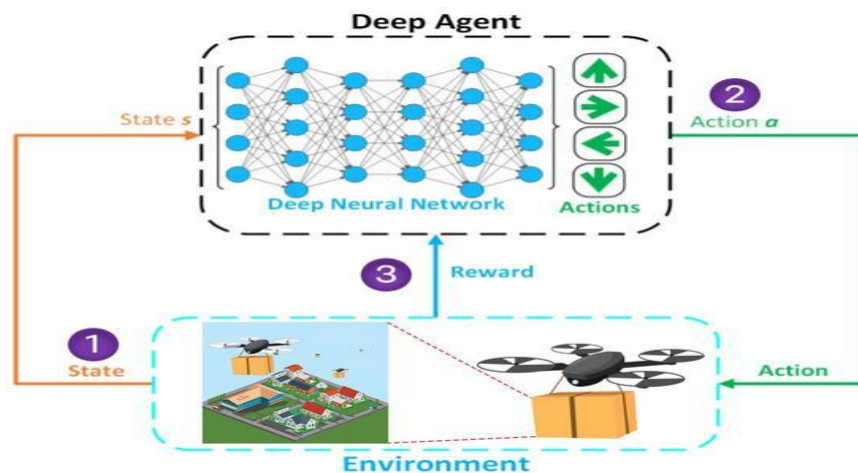
The state may contain:

- Current drone location
- Destination location
- Battery level
- Weather conditions
- Obstacles
- Traffic/airspace conditions
- Distance to destination

3. Actions

The drone can:

- Move forward
- Move backward
- Move left
- Move right
- Move upward
- Move downward
- Change route
- Hover/wait



4. Exploration

Exploration means trying new routes that the drone has not used before.

Example:

Known Route $\rightarrow$ Safe but longer

New Route $\rightarrow$ Unknown but potentially shorter

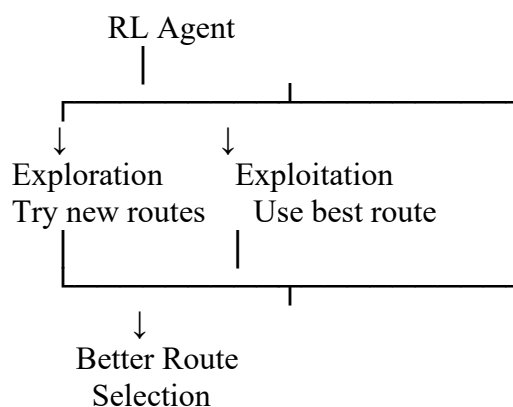
Exploration helps discover better routes and prevents the drone from always using the same route.

5. Exploitation

Exploitation means selecting the route that the RL agent already knows gives a high reward.

For example, if Route A consistently saves battery and time, the drone will prefer Route A.

Exploration vs Exploitation

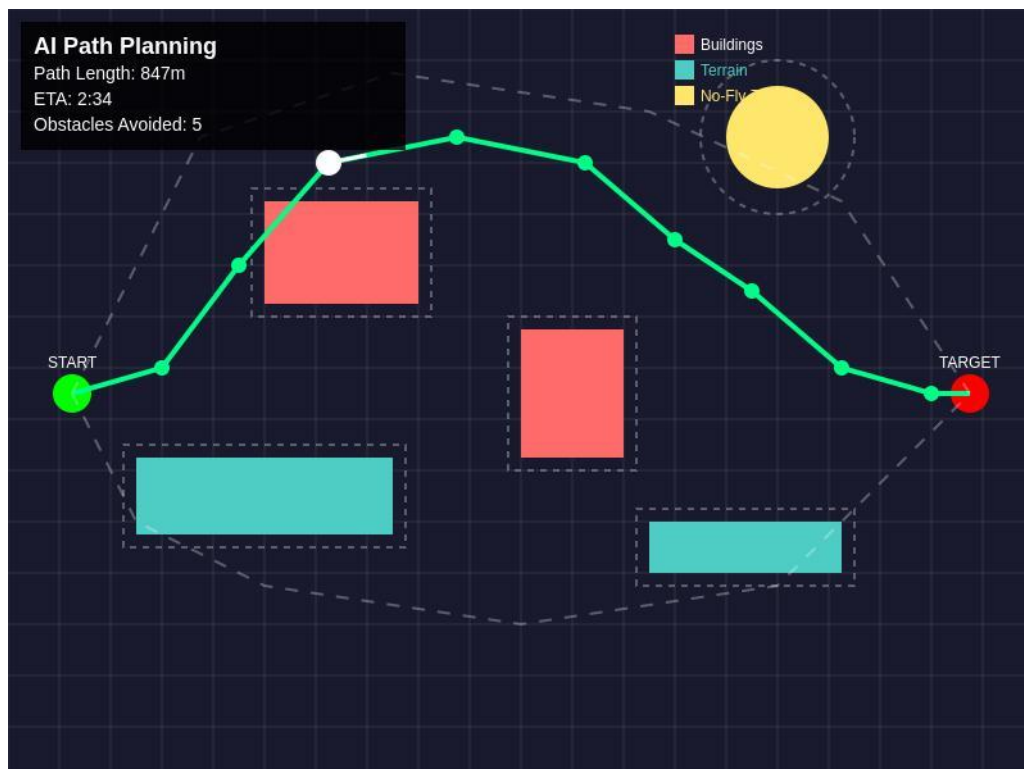




6. Reward Function

A possible reward system:

- Successful delivery $\rightarrow +100$
- Short route $\rightarrow +20$
- Low energy consumption $\rightarrow +10$
- Collision $\rightarrow -100$
- Excessive delay $\rightarrow -20$



Conclusion: Exploration helps the drone discover new and potentially better routes, while exploitation allows it to use routes already known to perform well. Balancing both improves navigation efficiency.

3.Design and develop a Reinforcement Learning framework a manufacturing robot uses Reinforcement Learning to optimize assembly operations. Apply the concept of policy and explain how value functions help improve efficiency.

Manufacturing Robot using RL

1. Objective

The objective is to make a manufacturing robot perform assembly operations with **minimum time, minimum errors, and efficient resource usage.**



2. State

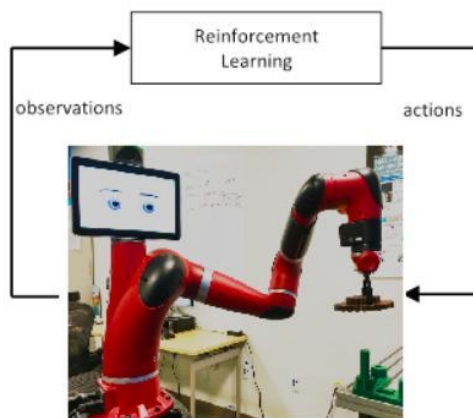
The state can include:

- Robot position
- Position of components
- Component availability
- Assembly stage
- Tool status
- Remaining time
- Previous operation

3. Actions

The robot can:

- Pick a component
- Move component
- Place component
- Tighten a screw
- Change tool
- Inspect component
- Repeat an operation



4. Policy

A **policy** defines how the RL agent selects an action for a given state.

Mathematically:

$\pi(a|s)$ = probability of selecting action **a** in state **s**

Example:

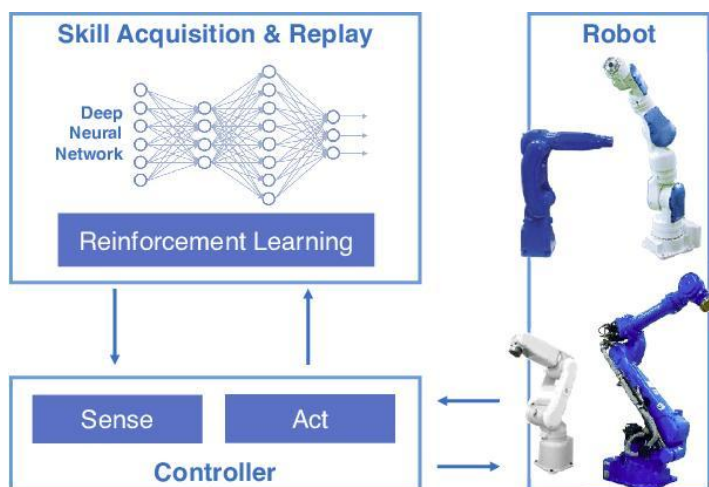
State: Component is available

↓
Policy

↓
Pick component

↓
Move to assembly

↓
Place component



5. Value Function

A value function estimates **how good a particular state is** in terms of future rewards.

For example:

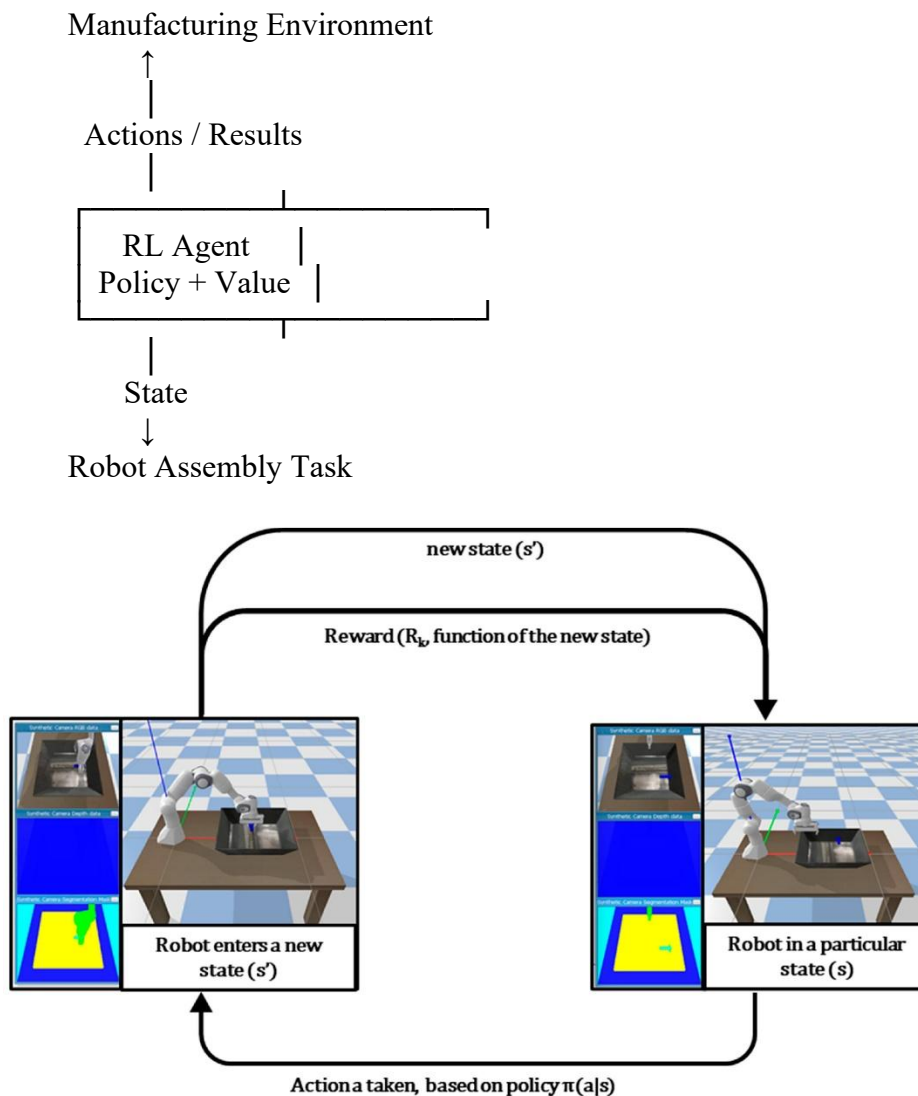
State A $\rightarrow$ Robot positioned correctly $\rightarrow$ High Value
State B $\rightarrow$ Robot far from component $\rightarrow$ Low Value

The robot prefers actions that lead to states with higher expected future rewards.

6. Reward

- Correct assembly $\rightarrow +50$
- Fast operation $\rightarrow +20$
- Successful inspection $\rightarrow +20$
- Assembly error $\rightarrow -50$
- Wasted movement $\rightarrow -10$
- Machine collision $\rightarrow -100$

Simple Framework



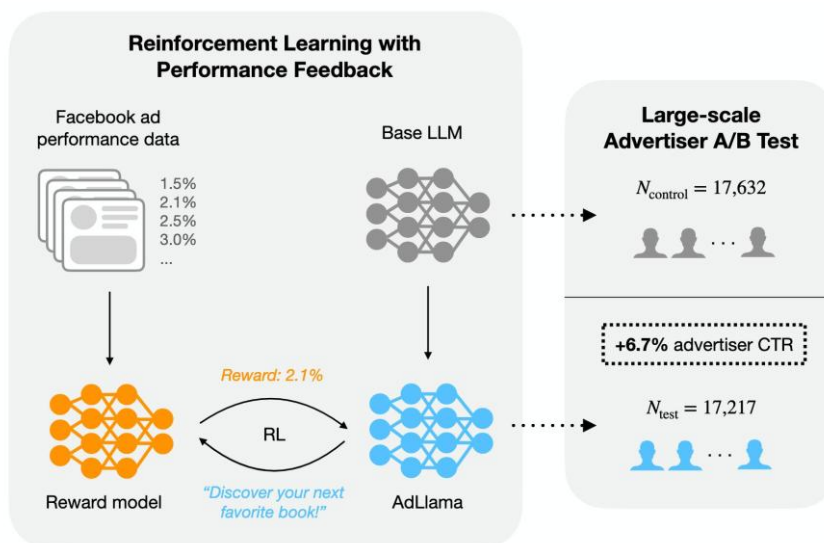
Conclusion: The policy decides what the robot should do, while the value function estimates the long-term benefit of actions. Together they help the robot reduce assembly time, errors, and unnecessary movements.

4. An online advertising system uses Reinforcement Learning to maximize click-through rates. Outline reward design challenges and explain their impact on system performance.

Online Advertising using Reinforcement Learning

1. Objective

The objective is to select advertisements that are most likely to receive **user clicks and engagement** while maintaining a good user experience.



2. State

The state may contain:

- User interests
- Previous clicks
- Browsing history
- Current webpage
- Advertisement type
- Time of day
- Device information

3. Actions

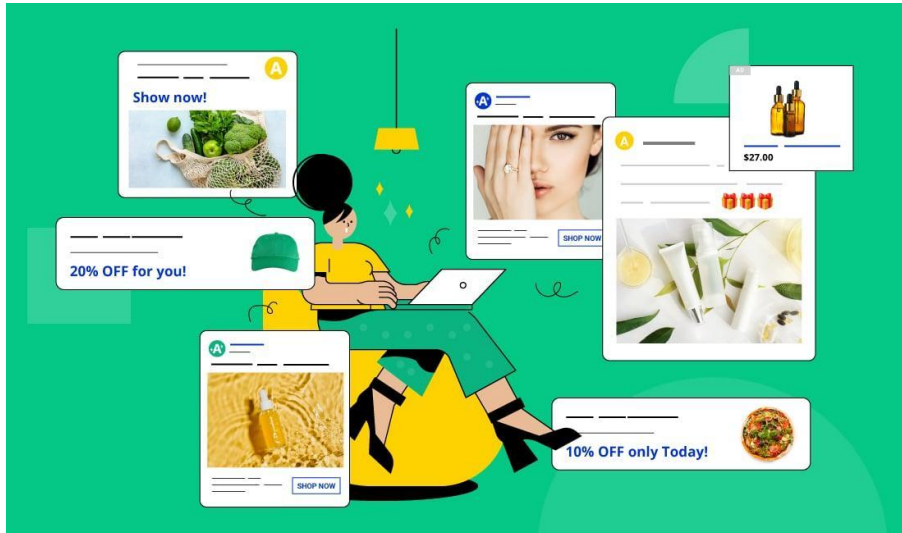
The RL agent can:

- Display Advertisement A
- Display Advertisement B
- Display Advertisement C
- Change advertisement
- Avoid showing an advertisement

4. Reward Design

A simple reward function could be:

Advertisement clicked → +10
Advertisement viewed → +2
Advertisement ignored → 0
User closes advertisement → -5
User reports advertisement → -10



5. Reward Design Challenges

Challenge 1: Clickbait

An advertisement may receive clicks but provide poor user experience.

Challenge 2: Short-term vs long-term reward

Maximizing immediate clicks may reduce long-term user satisfaction.

Challenge 3: Repeated advertisements

Showing the same advertisement repeatedly may increase short-term clicks but annoy users.

Challenge 4: User diversity

Different users have different interests, so the same reward strategy may not work for everyone.

6. Impact on Performance

Poor Reward Design



Incorrect Learning



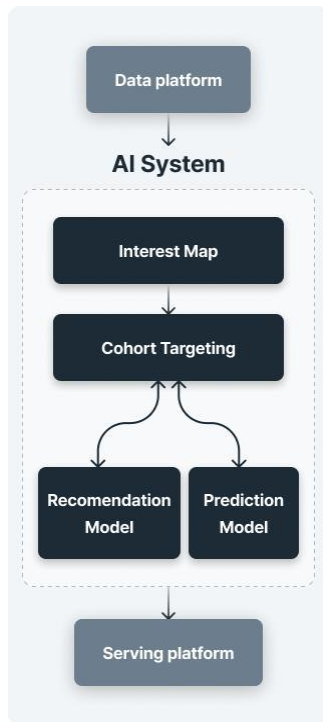
Poor Advertisement Selection



Low User Satisfaction



Poor Long-Term Performance



Conclusion

Reward design is very important because the RL agent learns directly from rewards. A good reward should consider not only **click-through rate**, but also user satisfaction, relevance, and long-term engagement.

5.A smart irrigation system uses Reinforcement Learning to optimize water usage. Create an RL framework by defining states, actions, and reward function.

Smart Irrigation System using Reinforcement Learning

1. Objective

The objective is to provide **the right amount of water at the right time** while maintaining healthy crops and reducing water wastage.



2. States

The RL state can include:

- Soil moisture
- Temperature
- Humidity
- Rainfall prediction
- Crop growth stage
- Water tank level
- Previous irrigation amount

Example:

Soil moisture = Low

Temperature = High

Rain probability = Low

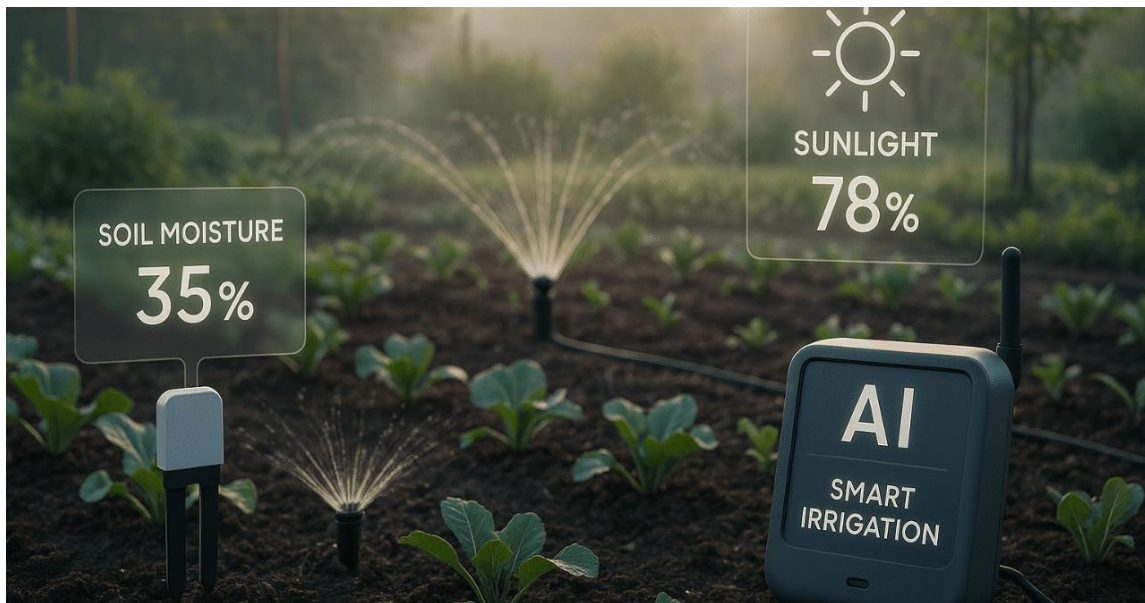
Crop stage = Flowering



3. Actions

The irrigation system can:

- Turn irrigation ON
- Turn irrigation OFF
- Use low water flow
- Use medium water flow
- Use high water flow
- Change irrigation duration



4. Reward Function

The reward should encourage healthy crops and reduced water consumption.

Healthy crop + Low water usage → +20

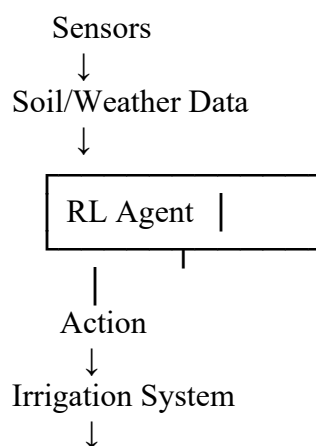
Adequate soil moisture → +10

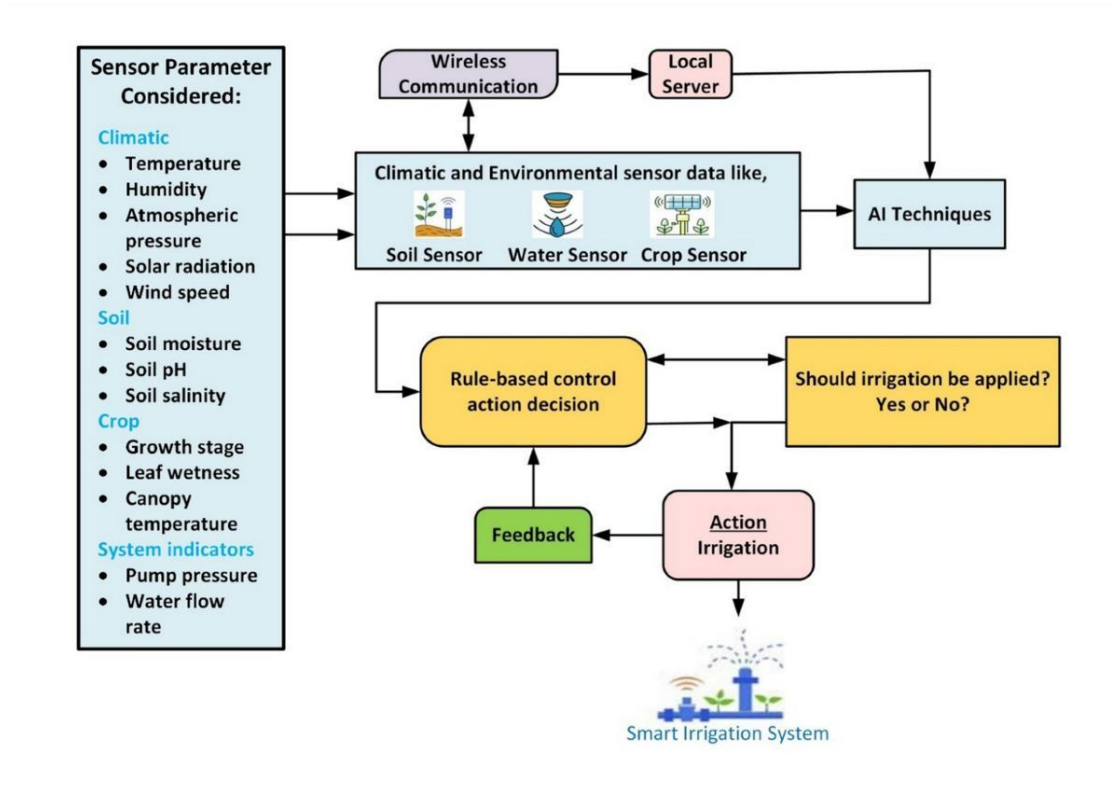
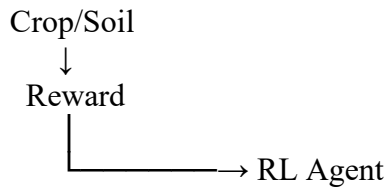
Excessive water usage → -10

Very dry soil → -20

Water wastage → -15

5. RL Framework





Conclusion

The RL agent continuously observes soil and environmental conditions and adjusts irrigation accordingly. This can reduce water wastage while maintaining suitable soil moisture for healthy crop growth.

Code of Conduct:

*I Malli Chandana(192472250) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Malli Chandana

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand